

Practice Quiz: Collective Intelligence — Module 04: Cognition

Name: ____ Date: ____

Part A: Multiple Choice

1. A **shared mental model** is: A) A document everyone has read B) The portion of the generative model that team members hold in common, enabling coordinated understanding C) A shared computer model D) A team mission statement
2. The Mann Gulch disaster illustrates: A) Poor firefighting equipment B) The catastrophic consequences of shared mental model collapse — when teams lose coordinated understanding, they disintegrate C) Insufficient training D) Bad weather
3. **Groupthink** is best understood as: A) Teams thinking together B) Excessive model alignment where cognitive diversity is suppressed — consensus prior has excessive precision C) Democratic decision-making D) Too many meetings
4. The optimal balance for team cognition is: A) Maximum alignment on everything B) Maximum diversity on everything C) High cognitive diversity combined with shared process protocols for integrating different models D) No communication at all
5. A **pre-mortem** exercise helps team cognition by: A) Predicting failures before they happen, testing model assumptions B) Celebrating in advance C) Reducing meeting time D) Eliminating risk
6. **After-action reviews** support collective cognition by: A) Assigning blame B) Providing structured opportunities for the team to update its shared model based on experience C) Fulfilling compliance requirements D) Generating reports
7. When a team member has a dissenting view, the value depends on: A) Their seniority B) Whether the dissent is based on different evidence or model structure, providing

information the consensus model lacks C) How loudly they argue D) Whether the manager agrees

Part B: Short Analysis

1. A product team consistently ships features that customers don't use. The team's internal alignment is strong — everyone agrees on what to build. Diagnose the cognitive failure: Where is the shared mental model wrong, and how would you update it?
2. Two sub-teams within a larger group have developed different interpretations of the project requirements. Rather than forcing agreement, design a process that leverages this cognitive diversity to produce a more robust shared understanding.
3. Your team has been operating effectively for 3 years. A new CEO announces a major strategic pivot. The team's shared mental model — built over years — is now partially obsolete. Design a process for rapidly updating the team's shared model to align with the new strategy while preserving the valid components.